SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA1613 COURSE CODE: Data Warehousing and Data Mining

---

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.
Assessment Tool Weightage: 20%

QUESTIONS: 05 **Total Marks:50**

Annexure A

COMPARITIVE ANALYSIS

Criteria	Scope of Items *(2)*	Comparison Depth(2.5)	Use of Evidence *(2)*	Critical Thinking *(2)*	Clarity of Presentation *(1)*	***Total(10)***
Weightag e	*20%*	25%	20%	20%	10%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

*I **G.MOHAMMAD SHAREEF/(192311288)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:G.MOHAMMAD SHAREEF

RUBRICS FOR CALCULATION:

		Comparative analysis			
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A

Q1 . Dataset: Assessment Tool 2: Comparative Analysis (25% → 5 Marks)

Time duration :45 Minutes
Assessment Tool 3 (Low)

Q1 . Dataset: Customer Data **Customer Income (₹) Spending Score**

C1	30000	60
C2	60000	40
C3	50000	70
C4	28000	65
C5	65000	35

Question:
Compare K-means and hierarchical clustering using this dataset and explain differences in performance and scalability. *(BL3 – 1 Mark)*

Q2. Dataset: Document Data

Document	Word1	Word2	Word3
D1	0.2	0.5	0.3
D2	0.1	0.6	0.3
D3	0.8	0.1	0.1
D4	0.75	0.15	0.1
D5	0.2	0.4	0.4

Question:
Compare K-means and EM algorithms in handling overlapping clusters for the given dataset. *(BL4 – 1 Mark)*

Q3. Dataset: Geographic Data

Point	Latitude	Longitude
P1	13.08	80.27
P2	13.10	80.25
P3	12.97	80.22
P4	13.05	80.30
P5	13.00	80.20

Question:
Compare Euclidean and Manhattan distance metrics and analyze their impact on clustering results. *(BL4 – 1 Mark)*

Q4. Dataset: Retail Products

Product	Sales (₹)	Frequency
P1	50000	20
P2	30000	10
P3	52000	22
P4	25000	8
P5	32000	12

Question:

Compare agglomerative and divisive clustering approaches and explain their suitability for this dataset. *(BL3 – 1 Mark)*

Q5 . Dataset: Mixed Attributes

Item	Price (₹)	Category	Rating
I1	500	A	4.5
I2	300	B	3.8
I3	550	A	4.7
I4	250	B	3.5
I5	520	A	4.6

Question:

Compare centroid-based and density-based clustering techniques and analyze their effectiveness on mixed data types. *(BL4 – 1 Mark)*

Q1. Compare K-Means and Hierarchical Clustering

Given Dataset

Customer	Income (₹)	Spending Score
C1	30,000	60
C2	60,000	40
C3	50,000	70
C4	28,000	65
C5	65,000	35

K-Means Clustering:

K-Means is a **partition-based clustering algorithm** that divides data into a predefined number of groups called clusters. Each cluster is represented by a **centroid**, which is the mean of the points belonging to that cluster.

Formula:

The commonly used Euclidean distance is:

$$d = \sqrt{(x_1 - c_1)^2 + (x_2 - c_2)^2}$$

The centroid is calculated as:

$$\mu = \frac{\sum x_i}{n}$$

K-Means attempts to minimize the **Within-Cluster Sum of Squares (WCSS)**:

$$WCSS = \sum_{j=1}^{K} \sum_{x_i \in C_j} ||x_i - \mu_j||^2$$

Step 1: Select K

Suppose: K = 2

So, the customers are divided into two groups.

Step 2: Select initial centroids

Assume two initial centroids are selected from the dataset.

Step 3: Calculate distance

For example, take C1 = (30000,60) and centroid = (28000,65).

$$
\begin{aligned}
d &= \sqrt{(30000-28000)^2 + (60-65)^2} \\
&= \sqrt{2000^2 + (-5)^2} \\
&= \sqrt{4,000,025} \\
&\approx 2000.006
\end{aligned}
$$

The same calculation is performed for all customers.

Step 4: Assign customers

Each customer is assigned to the closest centroid.

Customers such as C1 and C4 are similar because they have relatively low income and high spending scores. C2 and C5 are also similar because they have high income and relatively low spending scores.

Step 5: Recalculate centroid

For example, if C1 and C4 form a cluster:

For example, if C1 and C4 form a cluster:

$$Income = \frac{30000 + 28000}{2} = 29000$$

$$Score = \frac{60 + 65}{2} = 62.5$$

New centroid:

$$(29000, 62.5)$$

Step 6: Repeat the assignment and centroid calculation until the clusters become stable.

Hierarchical Clustering

Hierarchical clustering creates a **tree-like hierarchy of clusters**. Agglomerative hierarchical clustering starts with each customer as an individual cluster and gradually merges similar clusters.

Steps

1. Consider every customer as a separate cluster.

2. Calculate the distance between customers.
3. Find the closest two clusters.
4. Merge them.
5. Recalculate distances.
6. Continue until all customers form one hierarchy.
7. Use a **dendrogram** to select the required number of clusters.

and then further merge depending on the distance and linkage method.

Performance and Scalability Comparison:

K-Means	Hierarchical
Centroid-based	Hierarchy-based
Requires K initially	K can be selected from dendrogram
Faster	Usually slower
Good scalability	Less scalable
Suitable for large datasets	Suitable for small datasets
Produces clusters	Produces hierarchy/dendrogram

Conclusion:

For this small customer dataset, both methods can be used. **Hierarchical clustering** gives a better view of relationships between customers, while **K-Means is more suitable for large customer datasets because of its speed and scalability.**

Q2. Compare K-Means and EM algorithms in handling overlapping clusters for the given dataset.

Given Dataset

Document	Word1	Word2	Word3
D1	0.2	0.5	0.3
D2	0.1	0.6	0.3
D3	0.8	0.1	0.1
D4	0.75	0.15	0.1
D5	0.2	0.4	0.4

K-Means:

K-Means assigns each document to **one cluster** based on its distance from the centroid.

Formula:

$$d = \sqrt{(x_1 - c_1)^2 + (x_2 - c_2)^2 + (x_3 - c_3)^2}$$

or D1 and D2:

$$D1 = (0.2, 0.5, 0.3)$$

$$D2 = (0.1, 0.6, 0.3)$$

$$d = \sqrt{(0.2 - 0.1)^2 + (0.5 - 0.6)^2 + (0.3 - 0.3)^2}$$

$$= \sqrt{0.01 + 0.01}$$

$$\boxed{d = 0.1414}$$

Therefore, D1 and D2 are similar.

Similarly, D3 and D4 have very similar values and are likely to be grouped together.

K-Means Steps

1. Select .

2. Select initial centroids.
3. Calculate distances.
4. Assign documents to nearest centroid.
5. Recalculate centroids.
6. Repeat until convergence.

EM Algorithm

EM stands for **Expectation-Maximization**. Unlike K-Means, EM can assign a document to clusters using **probabilities**, making it more suitable for overlapping clusters.

Probability formula:

$$P(C_k|x) = \frac{P(x|C_k)P(C_k)}{\sum_j P(x|C_j)P(C_j)}$$

EM Steps:

1. Initialize cluster parameters.
2. **E-Step:** Calculate probability of each document belonging to each cluster.
3. **M-Step:** Update cluster parameters.
4. Repeat E-Step and M-Step until convergence.

For example, a document could have:

$$P(C_1|D5) = 0.6$$

$$P(C_2|D5) = 0.4$$

This means D5 has characteristics of both clusters.

Comparison:

K-Means	EM
Hard clustering	Soft clustering
One cluster membership	Probability of multiple memberships
Distance-based	Probability-based
Simple and faster	More computationally complex
Less suitable for overlapping clusters	Better for overlapping clusters

Conclusion

For the given document dataset, **K-Means works well when documents are clearly separated**, while **EM is more suitable when documents have overlapping characteristics and may belong partially to more than one cluster**.

Q3. Compare Euclidean and Manhattan distance metrics and analyze their impact on clustering results.

Given Dataset

Point	Latitude	Longitude
P1	13.08	80.27
P2	13.10	80.25
P3	12.97	80.22
P4	13.05	80.30
P5	13.00	80.20

Euclidean Distance

Euclidean distance represents the straight-line distance between two points.

Formula:

$$D_E = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

For P1 and P2:

$$P1 = (13.08, 80.27)$$

$$P2 = (13.10, 80.25)$$

Substitution:

$$D_E = \sqrt{(13.10 - 13.08)^2 + (80.25 - 80.27)^2}$$

$$= \sqrt{(0.02)^2 + (-0.02)^2}$$

$$= \sqrt{0.0004 + 0.0004}$$

$$\boxed{D_E = 0.0283}$$

Manhattan Distance

Manhattan distance calculates the sum of absolute differences.

Formula:

$$D_M = |x_2 - x_1| + |y_2 - y_1|$$

or P1 and P2:

$$D_M = |13.10 - 13.08| + |80.25 - 80.27|$$

$$= 0.02 + 0.02$$

$$\boxed{D_M = 0.04}$$

Impact on Clustering:

The two methods give different distance values.

- Euclidean: 0.0283
- Manhattan: 0.04

Therefore, the choice of distance metric can change which points are considered closest.

This can affect:

- Cluster formation
- Cluster membership
- Cluster boundaries
- Final clustering result

Comparison:

Euclidean	**Manhattan**
Straight-line distance	Sum of coordinate differences
Uses square root	No square root
More affected by large differences	Less affected by individual large differences
Common in geometric clustering	Useful for grid/high-dimensional situations

Conclusion:

For geographic numerical data, Euclidean distance gives direct geometric similarity, while Manhattan distance measures coordinate-wise differences. **Choosing the appropriate metric is important because it can change the final clusters.**

Q4. Compare agglomerative and divisive clustering approaches and explain their suitability for this dataset.

Given Dataset

Product	Sales (₹)	Frequency
P1	50,000	20
P2	30,000	10
P3	52,000	22
P4	25,000	8
P5	32,000	12

Agglomerative Clustering

Agglomerative clustering follows a **bottom-up approach**.

Formula:

Euclidean distance:

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

For P1 and P3:

$$P1 = (50000, 20)$$

$$P3 = (52000, 22)$$

$$d = \sqrt{(52000 - 50000)^2 + (22 - 20)^2}$$

$$= \sqrt{2000^2 + 2^2}$$

$$= \sqrt{4000004}$$

$$\boxed{d \approx 2000.001}$$

Thus, P1 and P3 are quite similar compared with products having much larger differences.

Steps:

1. Start with every product as an individual cluster.
2. Calculate distances.

3. Find the closest products/clusters.
4. Merge them.
5. Recalculate cluster distances.
6. Continue merging.
7. Create a dendrogram.
8. Select the required clusters.

Divisive Clustering:

Divisive clustering follows a **top-down approach**.

Steps

1. Start with all products in one cluster.
2. Divide the cluster into smaller groups.
3. Identify groups that need further splitting.
4. Continue splitting.
5. Stop when the required number of clusters is obtained.

Comparison:

Agglomerative	**Divisive**
Bottom-up	Top-down
Starts with individual clusters	Starts with one cluster
Merges clusters	Splits clusters
Easier for small datasets	More computationally demanding
Builds hierarchy	Builds hierarchy

Suitability:

This dataset contains only **five products**, so both approaches can be applied.

Agglomerative clustering is more convenient because it can easily identify similar products such as:

$$P1 \approx P3$$

and

$$P2 \approx P5$$

based on their sales and frequency.

Conclusion:

Agglomerative clustering is more suitable for this small retail dataset because it is simple and clearly shows the similarity between products.

Q5. Compare centroid-based and density-based clustering techniques and analyze their effectiveness on mixed data types.

Given Dataset

Item	Price (₹)	Category	Rating
I1	500	A	4.5
I2	300	B	3.8
I3	550	A	4.7
I4	250	B	3.5
I5	520	A	4.6

The dataset contains both **numerical and categorical data**.

- Price → Numerical
- Rating → Numerical
- Category → Categorical

Centroid-Based Clustering:

K-Means is an example of centroid-based clustering.

It groups items around a central point called a **centroid**.

Formula:

$$WCSS = \sum_{j=1}^{K} \sum_{x_i \in C_j} ||x_i - \mu_j||^2$$

Steps:

1. Select K.
2. Select initial centroids.
3. Calculate distances.
4. Assign items to nearest centroid.
5. Calculate new centroids.
6. Repeat until convergence.

In this dataset, I1, I3 and I5 have:

- Category A
- High prices
- High ratings

Therefore, they show strong similarity.

I2 and I4 have:

- Category B
- Lower prices
- Lower ratings

So they can form another group.

Density-Based Clustering:

DBSCAN is an example of density-based clustering.

It forms clusters by identifying regions containing a high concentration of points.

Parameters:

$$\epsilon = \text{neighborhood radius}$$

$$MinPts = \text{minimum number of points}$$

Core Point Formula:

$$|N_\epsilon(p)| \geq MinPts$$

Steps:

1. Select ε.
2. Select MinPts.
3. Find neighboring points.
4. Identify core points.
5. Expand dense clusters.
6. Mark isolated points as noise.

Effectiveness on Mixed Data:

The main problem is that **Category A and Category B are categorical values** and cannot directly be used in ordinary Euclidean distance.

A suitable mixed-data distance can be represented as:

$$D = w_1 D_{numeric} + w_2 D_{category}$$

For categorical data:

$$D_{category} = \begin{cases} 0, & \text{same category} \\ 1, & \text{different category} \end{cases}$$

Therefore, preprocessing or an appropriate mixed-data clustering technique is required.

Comparison:

Centroid-Based	**Density-Based**
K-Means	DBSCAN
Uses centroids	Uses density
K is generally specified	K is not required
Good for compact clusters	Good for irregular clusters
Sensitive to outliers	Can detect noise
Needs numerical representation	Needs suitable distance for mixed data

Conclusion:

For the given dataset, the items naturally show two groups: **I1, I3, I5** and **I2, I4**. Centroid-based clustering can identify these groups after suitable encoding/preprocessing. Density-based clustering can be useful when the data contains irregular clusters or noise. For mixed data, the **choice of representation and distance measure is very important**.